

RF System and Amplifier Design using Keysight ADS

System-Level RF Characterization and Low-Noise Amplifier Design Portfolio

Topics: ACPR/ACLR · EVM · Receiver Link Budget · Spurious Response · IP3/TOI · Transistor Biasing · S-Parameters · Noise Figure · Stability · Smith Chart Matching · Input/Output L-C Matching

Tools: Keysight Advanced Design System (ADS)

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Abstract

This report presents a two-part radio-frequency (RF) design study carried out in Keysight Advanced Design System (ADS). The first part characterizes an RF amplifier and receiver chain at the system level, quantifying linearity and signal quality through adjacent-channel power ratio (ACPR) and error-vector magnitude (EVM) under a $\pi/4$ -DQPSK modulated source, followed by a cascaded receiver link-budget, spurious-response mapping, and two-tone third-order intercept (IP3) analysis with and without intermediate-frequency (IF) filtering. The second part addresses the circuit level: a single-transistor amplifier is biased, its device model is verified against measured S-parameters, its noise figure and stability are analyzed using source-pull noise circles, and input/output impedance-matching networks are synthesized. Together the two parts traverse the complete RF design flow, from system specification and modulated-signal linearity down to a biased, model-verified, noise-optimized, and impedance-matched amplifier.

The project demonstrates a complete RF design workflow: define system metrics, simulate nonlinear and modulated behavior, extract gain/noise/intermodulation results, bias a transistor, verify the device model, select source and load matching networks, and summarize the engineering trade-offs in a defensible way.

Index Terms: ACPR, EVM, error-vector magnitude, link-budget, low-noise amplifier (LNA), noise figure, third-order intercept (IP3), S-parameters, impedance matching, Keysight ADS.

Design area	Method / metric	Outcome
Transmitter linearity	ACPR / ACLR with $\pi/4$ -DQPSK	Adjacent-channel spectral regrowth evaluated; more negative ACPR indicates better leakage performance
Signal quality	EVM and constellation error	Ideal and distorted symbol trajectories compared; RMS EVM equation defined.
Receiver chain	Cascaded gain and noise figure	Approx. 76.1 dB gain and approx. 4.2 dB cascaded NF from visible ADS labels.
Interference	Spurious + two-tone IP3	Spurious products and IM3 terms identified; IF filtering suppresses post-generation out-of-band products.
Device design	Spurious products and IM3 terms identified; IF filtering suppresses post-generation out-of-band products.	Bias point selected and model checked before noise/matching design.
LNA design	Noise circles + stability + source match	NFmin approx. 1.744 near 1.86 GHz; Sopt and source-match tradeoff extracted.
Power transfer	Output L/C matching	Output network synthesized
Matching network	Input/output L-C matching and stabilization	Cin 30 pF, Lin 4 nH, Rstab 39 ohm, Lstab 25 nH, Lout 16 nH, Cout 2 pF

Power transfer	Input/output L-C matching	Stabilization and matching elements summarized from ADS screenshots.
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Table 1. Summary of the design tasks, methods, and outcomes.

1. Introduction and Design Flow

Modern RF front-ends face two simultaneous and competing requirements. They must be linear enough that a digitally-modulated signal is neither distorted within its own band nor allowed to regrow into adjacent channels, and they must be sensitive enough to recover weak desired signals against the noise floor while tolerating strong nearby interferers. The first requirement is governed by large-signal metrics ACPR, EVM, and the third-order intercept point; the second by the noise figure and the cascaded dynamic-range budget of the receiver.

This study uses Keysight ADS to address both, progressing from behavioural system models to a transistor-level circuit. The system-level work (Sections 2–6) establishes the linearity and budget of the chain; the circuit-level work (Sections 7–11) designs a physical transistor amplifier- biasing, model verification, noise and stability, and impedance matching.

Table 2 lists each design activity and its purpose.

Design activity	Purpose
ACPR analysis	Quantify adjacent-channel spectral regrowth from non-linearity
EVM analysis	Measure modulated-signal quality against the ideal constellation
Receiver link-budget	Cascade gain, noise figure, and intercept to predict sensitivity
Spurious-response analysis	Identify in-band mixing products that mimic signal
Third-order intercept (IP3)	Quantify two-tone intermodulation and dynamic range
Transistor bias setup	Fix the DC operating point of the amplifier
S-parameter model verification	Validate the device model against datasheet data
Noise & stability analysis	Determine NFmin, stability, and the optimum source match
Impedance matching	Synthesize input and output matching networks

Table 2a. Design activities and their purpose in the RF design flow.

Metric	Value
ACPR / ACLR	Peak adjacent-channel lobes approx. 50-60 dB below the main-lobe peak from spectrum plot
EVM	Peak error-vector magnitude approx. 0.011 normalized units; RMS EVM (%) not printed
Receiver cascade gain	Approx. 76.1 dB
Receiver cascaded NF	Approx. 4.2 dB
Two-tone test A	Pout fundamentals = 7.519 dBm; OIP3 = 21.828 / 21.734 dBm; IIP3 = -25.691 / -25.785 dBm
Two-tone test B	Pout fundamentals = 6.895 dBm; OIP3 = 21.658 / 21.507 dBm; IIP3 = -25.237 / -25.394 dBm
Bias operating point	IBB = 19.40; VBE = 774.4 mV; VCC current = -2.014 mA
Optimum source match	Sopt = 0.440 angle 98 deg at 2.0 GHz; normalized impedance, Z = 0.613 + j0.662
Input/output reflection	S11 = 0.560 angle -128 deg; S22 = 0.700 angle -43 deg at 2.0 GHz
Noise performance	NFmin = 1.744 at 1.86 GHz; Rn = 14.0 at 2.0 GHz
Small-signal gain	dB(S21) = 9.078 dB at 1.93 GHz
Final matching components	Cin=30 pF, Lin=4 nH, Rstab=39 ohm, Lstab=25 nH, Lout=16 nH, Cout=2 pF

Table 2b. Consolidated ADS Numerical Re

2. Adjacent-Channel Power Ratio (ACPR/ACLR)

ACPR, also called adjacent-channel leakage ratio (ACLR), is a transmitter linearity metric. It compares the integrated power leaking into an adjacent channel against the integrated power in the wanted channel. It is normally reported in dBc, so a more negative value indicates better adjacent-channel leakage performance.

$$ACPR_{dBc} = 10 \log_{10} (P_{adj} / P_{main}) \quad (1)$$

where P_{main} is the integrated power in the wanted channel and P_{adj} the integrated power in a neighbouring channel. ACPR is normally reported in dBc; therefore a more negative ACPR value indicates better adjacent-channel leakage performance. Spectral regrowth into the adjacent channel is a direct consequence of amplifier non-linearity acting on a non-constant-envelope signal. The test bench of Figure 1 drives an RF amplifier with a $\pi/4$ -DQPSK source and probes the upper, main, and lower channel powers; Figure 2 adds receive-side filtering so that ACPR can be evaluated including channel-select effects.

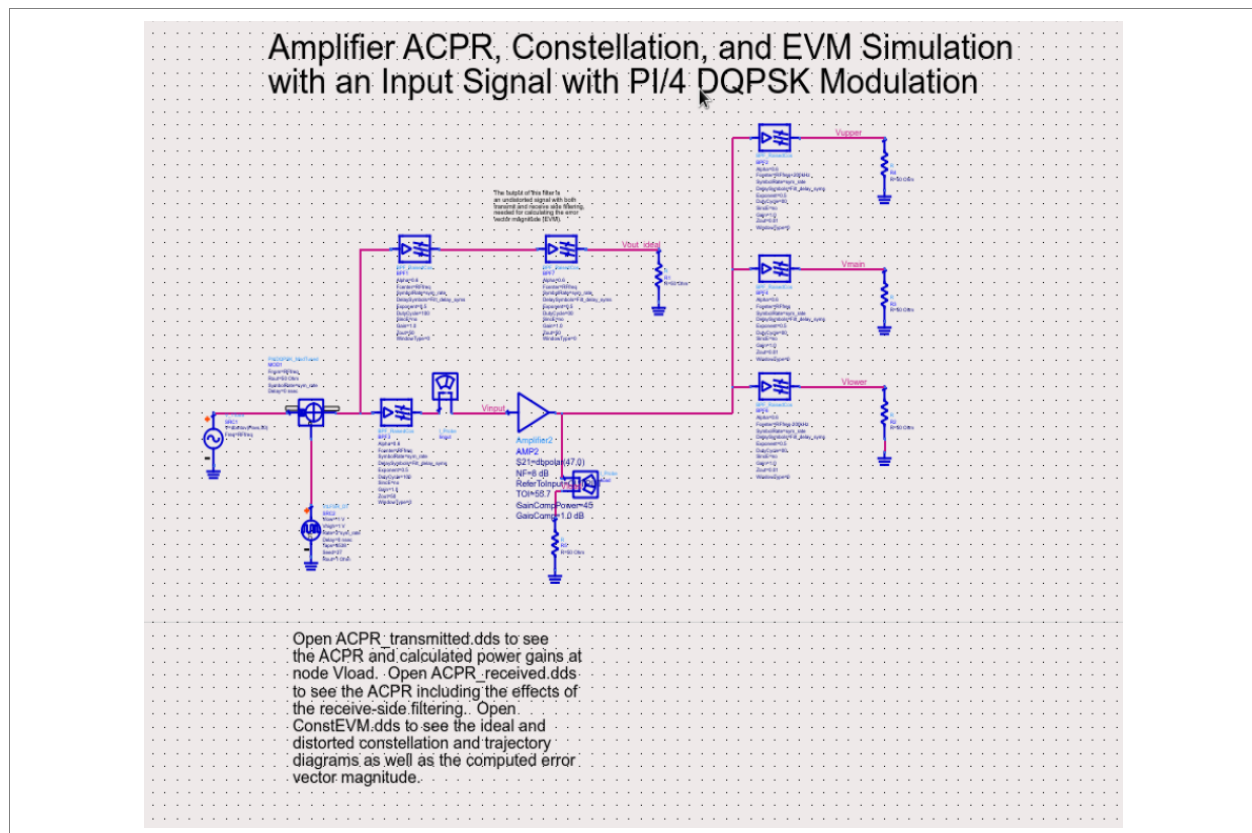


Figure 1. ACPR / constellation / EVM test bench.

The RF amplifier is driven by a $\pi/4$ -DQPSK source, and upper, main, and lower channel power probes are used for ACPR evaluation.

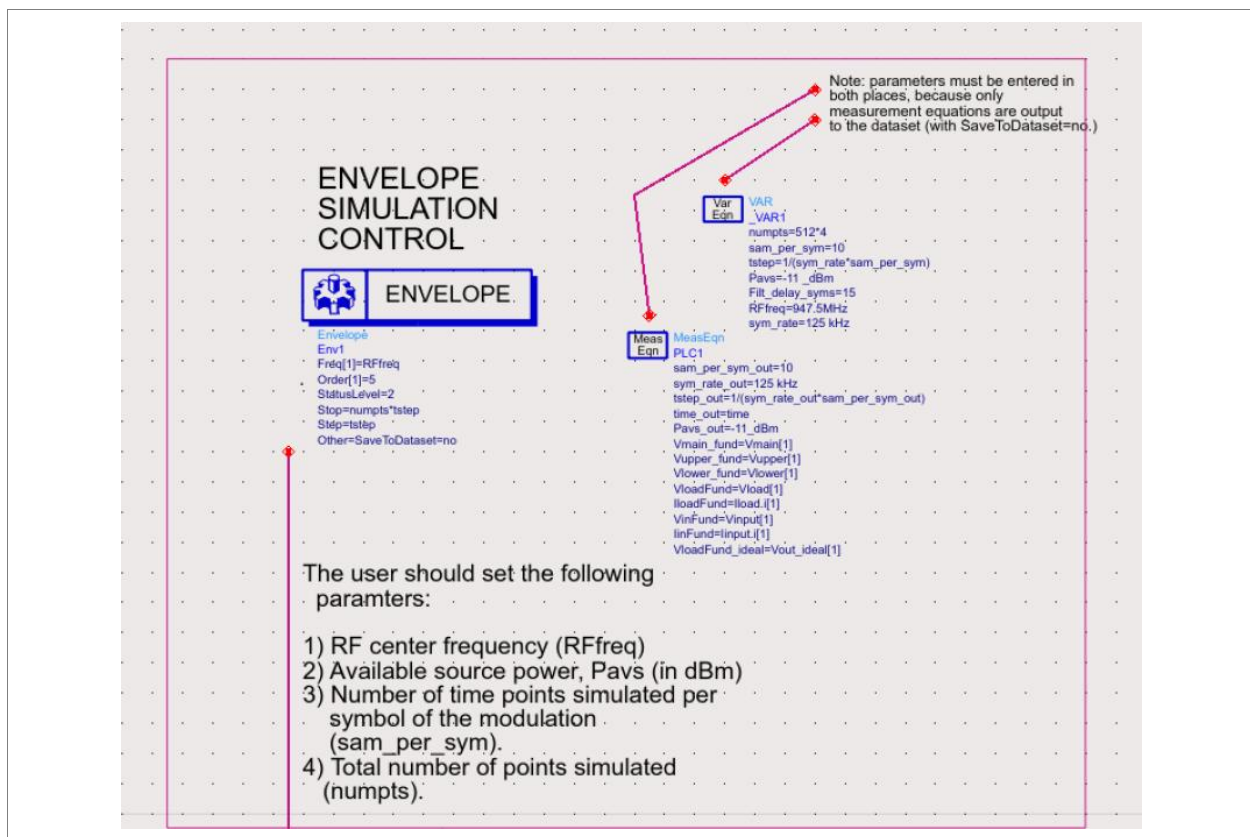


Figure 2. Envelope-simulation control and ACPR continuation block.

The setup defines RF center frequency, available source power, sampling per symbol, and simulation length for modulated-envelope analysis.

The transmitted spectrum and signal trajectory are shown in Figure 3. The regrowth lobes flanking the main channel are the spectral signature of third- and higher-order distortion; the trajectory diagram shows the path of the modulated signal in the I/Q plane.

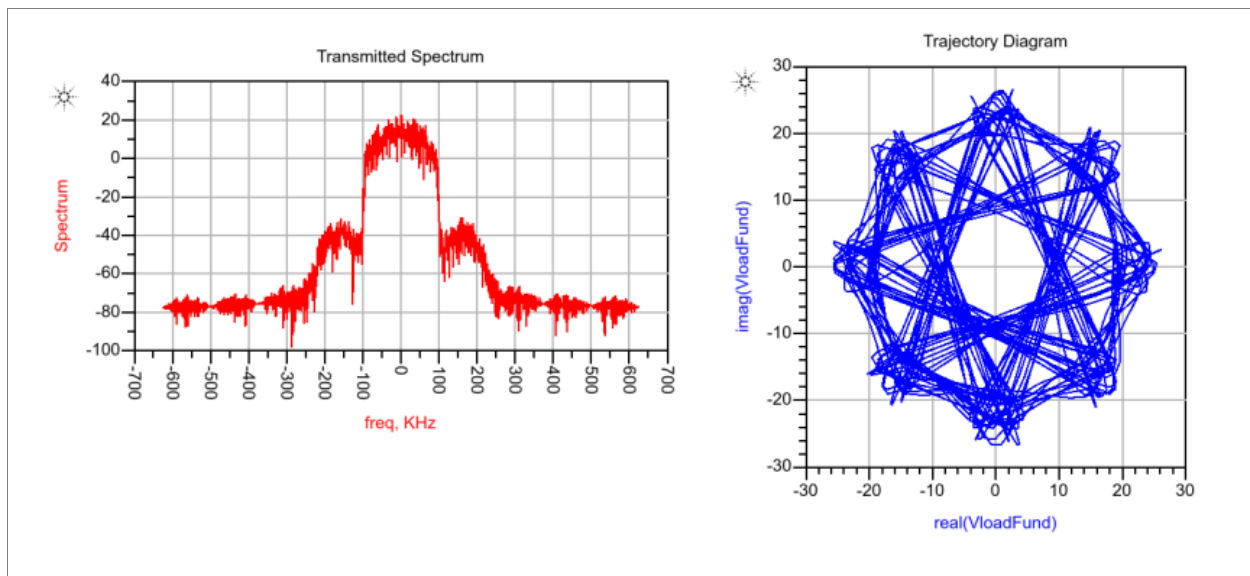


Figure 3. Transmitted spectrum and trajectory diagram.

The spectrum shows adjacent-channel regrowth around the main channel; the trajectory diagram shows the modulated I/Q signal path. The adjacent lobes visible in the spectrum are roughly 50-60 dB below the main-lobe peak.

3. Error-Vector Magnitude (EVM)

EVM evaluates the quality of the wanted modulated signal. For each symbol, the error vector is the difference between the measured symbol and the ideal reference symbol. RMS EVM normalizes the total squared error by the ideal reference symbol energy. Where ACPR looks outward at neighbouring channels, EVM looks inward at the wanted signal. For each received symbol, the error vector is the difference between the measured constellation point and its ideal location; EVM is the root-mean-square error-vector magnitude normalized to the reference signal power:

$$EVM_{rms} = \sqrt{(\text{sum}(|S_{meas,k} - S_{ideal,k}|^2) / \text{sum}(|S_{ideal,k}|^2))} \times 100\% \quad (2)$$

where $e_k = S_{meas,k} - S_{ideal,k}$ is the error vector of the k -th symbol and S_{ref} the reference magnitude. This normalized RMS form compares the measured constellation against the ideal reference constellation across all N symbols. EVM captures the combined effect of non-linearity, noise, phase error, amplitude imbalance, and filtering on the demodulated symbols, making it one of the most informative quality metrics for a modulated RF link. Figure 4 compares the distorted and ideal constellations; Figure 5 shows the schematic that derives EVM by differencing the ideal and amplifier paths.

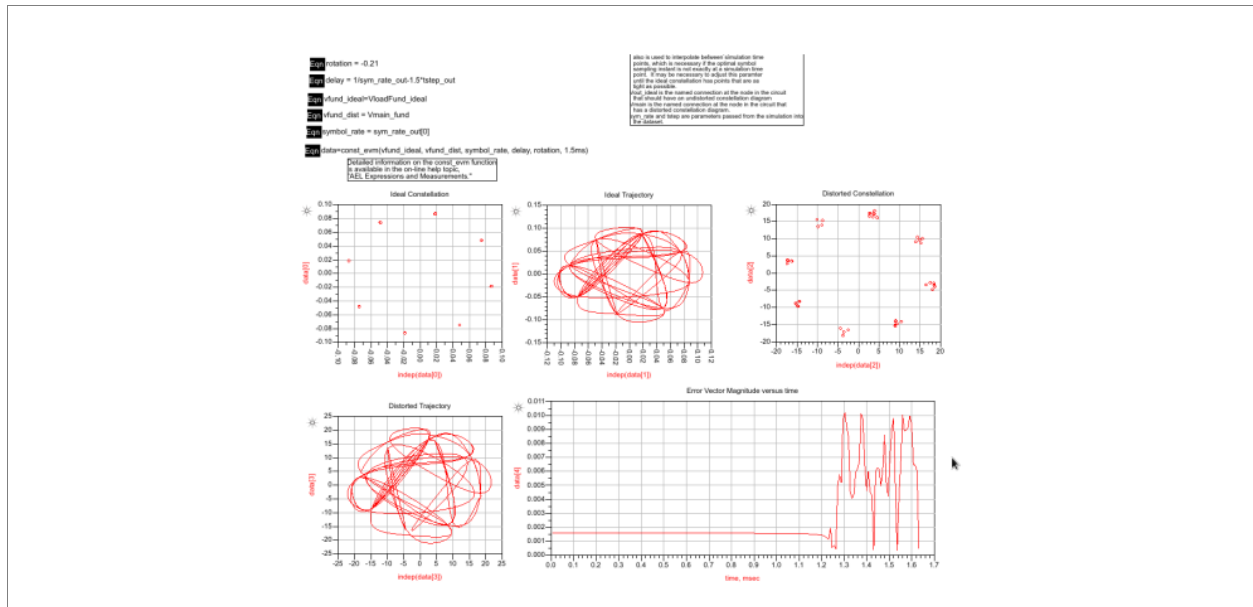


Figure 4. EVM evaluation- distorted versus ideal constellation.

The figure compares ideal and distorted constellations and shows the computed error-vector behavior.

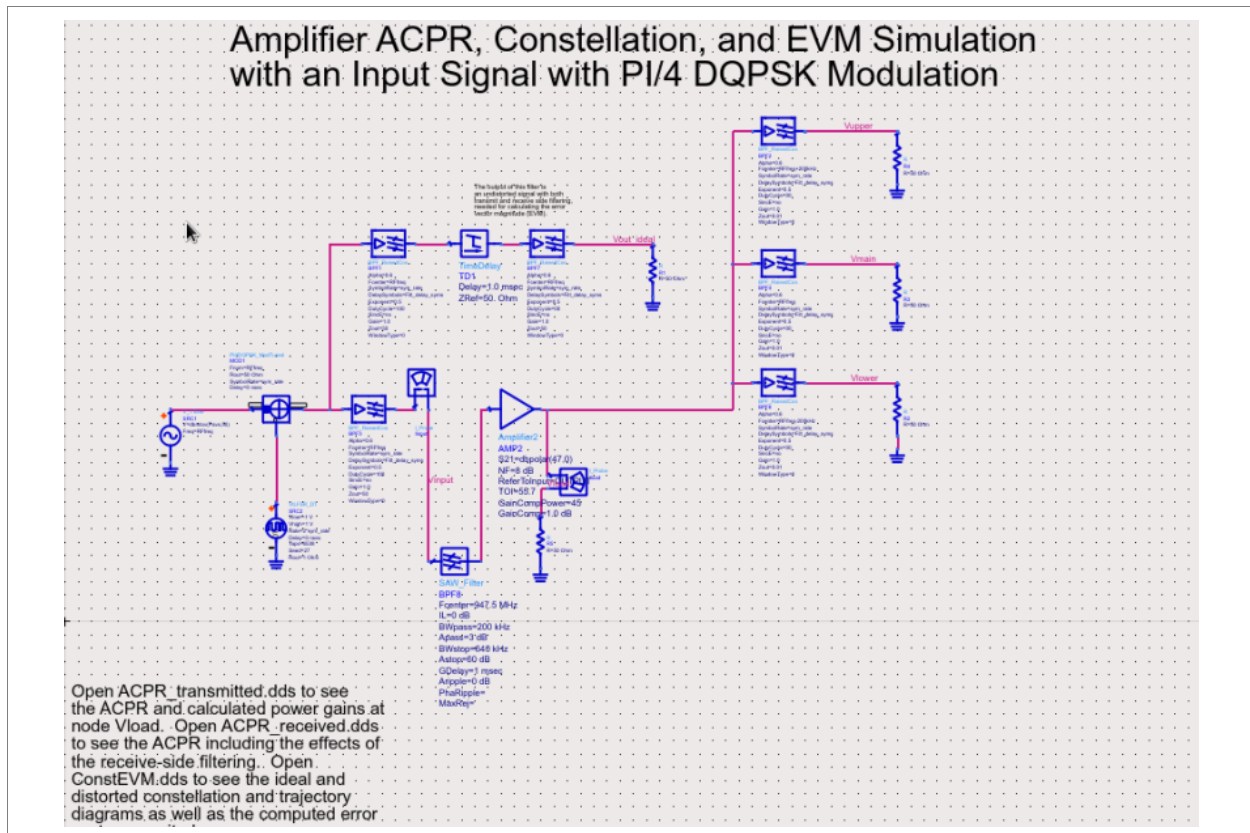


Figure 5. EVM schematic.

The ideal and distorted signal paths are compared to compute error-vector magnitude. A tightly clustered constellation corresponds to low EVM and a high-fidelity link; spreading of the symbol clouds indicates the onset of non-linear distortion and sets the practical input-power ceiling of the amplifier.

4. Receiver Link-Budget Analysis

A receiver budget cascades the gain, noise figure, and intercept of every stage to predict end-to-end sensitivity and dynamic range before any circuit is built. The cascaded noise figure follows Friis's formula:

$$F_{\text{tot}} = F_1 + (F_2 - 1)/G_1 + (F_3 - 1)/(G_1 G_2) + \dots \quad (3)$$

$$F = 10^{(\text{NFdB}/10)}, \quad G = 10^{(\text{GdB}/10)} \quad (3a)$$

Friis calculation uses linear noise factors and linear power gains. Noise figure and gain values expressed in dB must therefore be converted to linear scale before applying the cascade equation which shows why the first stage dominates the system noise figure and must therefore be a low-noise, high-gain element. The Friis equation must be evaluated using linear noise factors and linear gains, not dB values. From the visible ADS budget schematic, the approximate cascaded gain is 76.1 dB and the approximate cascaded noise figure is 4.2 dB. These values are calculated from the visible stage labels and should be replaced by the ADS budget-controller output if the raw ADS dataset is available. Figure 6 shows the per-stage specifications (gain, NF, and intercept) feeding the ADS budget controller.

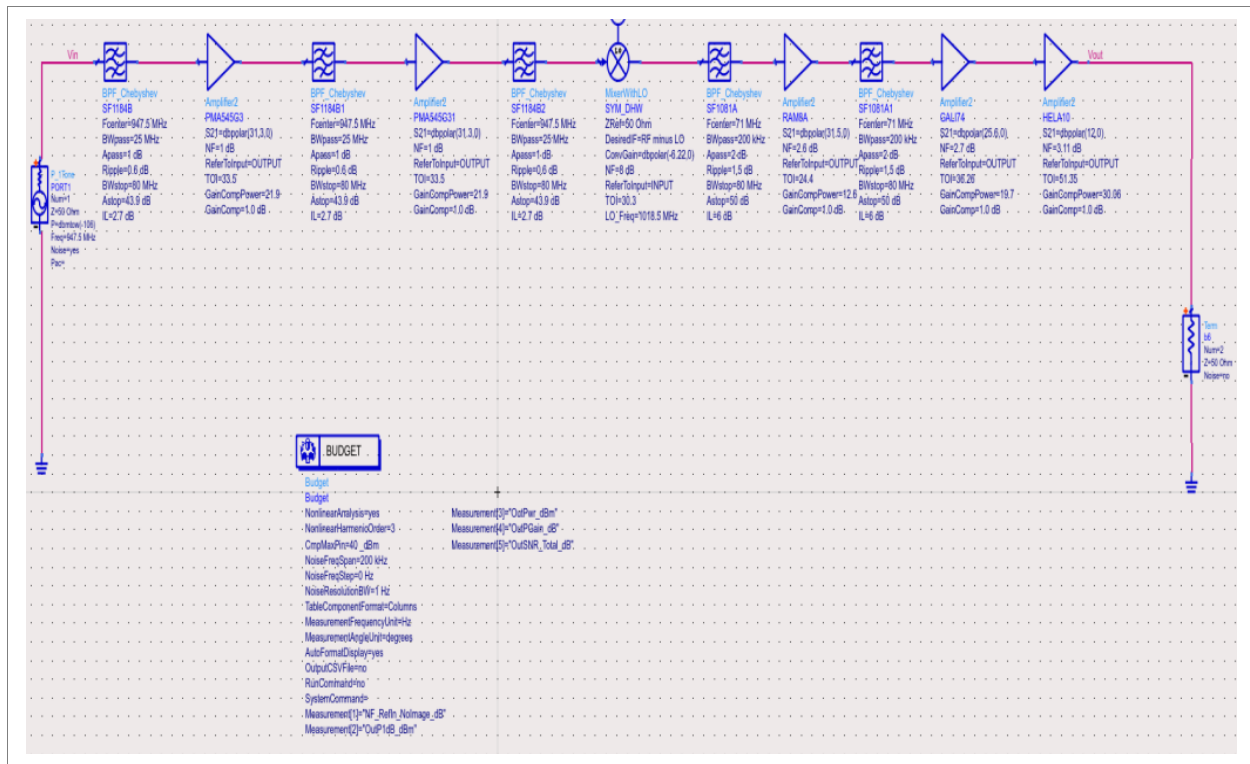


Figure 6. Receiver specifications for budget analysis.

The chain includes RF filtering, gain blocks, mixer conversion, IF filtering, and additional gain stages.

Budget analysis is where system feasibility is decided: noise figure is allocated to the front stages and linearity to the back stages, and the cascade is verified to close against the sensitivity and dynamic-range targets. The cascade includes RF filtering, gain blocks, a mixer, IF filtering and additional IF/baseband gain. The visible labels provide stage gain/loss, noise figure and intercept information.

From the visible ADS budget schematic, the approximate total receiver gain is 76.1 dB and the approximate cascaded noise figure is 4.2 dB. The first RF filter and first RF amplifier dominate the cascaded noise figure, which is consistent with Friis theory.

Budget result	Value	Interpretation
Approx. total receiver gain	76.1 dB	Sum of visible stage gain/loss values
Approx. cascaded NF	4.2 dB	Friis calculation using visible stage labels
Dominant NF contributor	First RF filter + first RF amplifier	Early stages dominate cascaded NF
System design implication	Low-loss RF front-end and early LNA gain are critical	Improves sensitivity before later-stage noise is added

Table 3. Budget result value with Interpretation

4. Spurious-Response and Intermodulation Analysis

Spurious-response analysis identifies unwanted products generated by nonlinear blocks such as mixers and amplifiers. Mixers and other non-linear stages generate spurious products - mixing tones at frequencies $m \cdot f_{RF} \pm n \cdot f_{LO}$ that can fall in-band and masquerade as signal. Mixer products appear at combinations of RF and LO frequencies and may fall into the wanted IF band if frequency planning and filtering are not sufficient.

$$f_{spur} = |m f_{RF} + /- n f_{LO}| \text{-----(4)}$$

Mapping these responses, as in Figure 7, identifies the frequencies at which the receiver is vulnerable to interference and informs filter placement. If a product falls into the IF bandwidth, it can appear as a false wanted signal and must be suppressed by filtering or frequency planning.

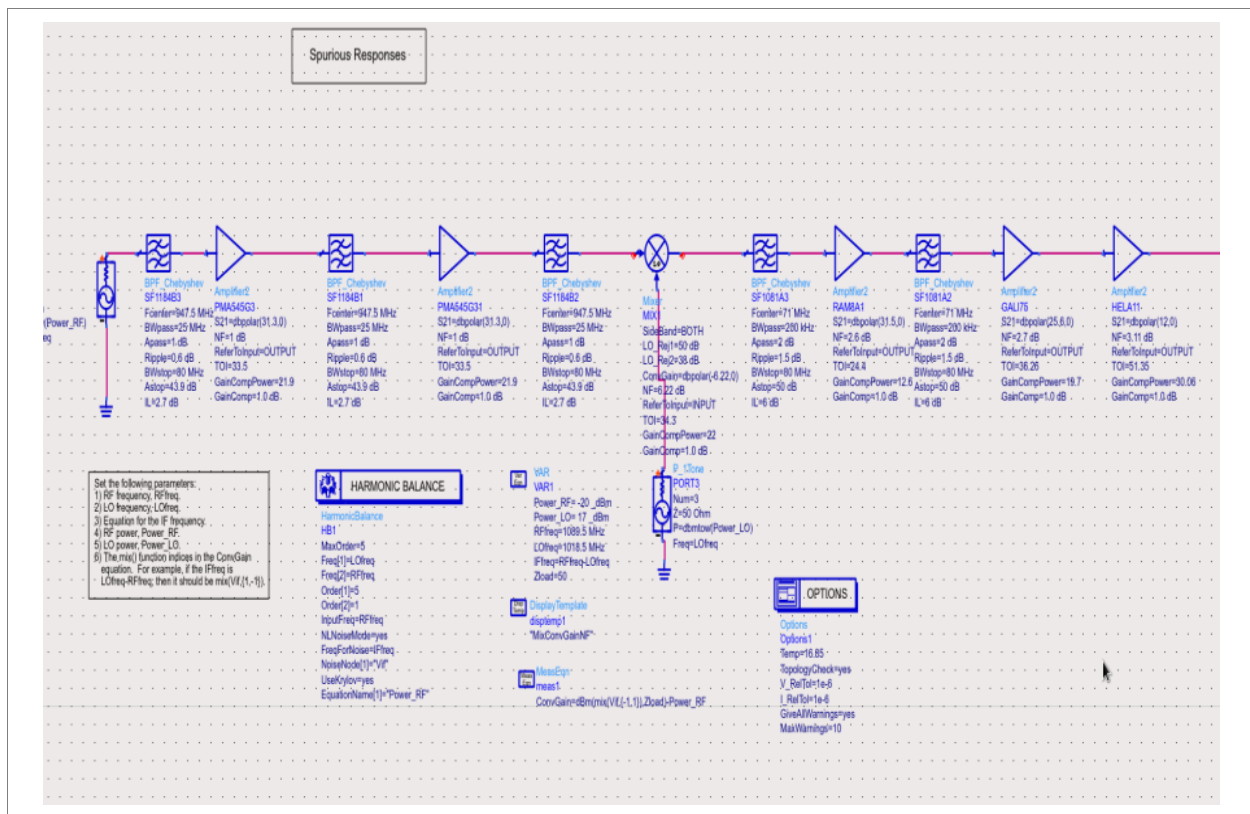


Figure 7. Schematic for spurious-response analysis.

The ADS harmonic-balance setup maps mixer and nonlinear-stage products across the receiver chain. This is technically appropriate for identifying interfering combinations of RF, LO and harmonic products.

Third-order intermodulation is particularly important because IM3 products appear close to the wanted tones. For two tones f_1 and f_2 , the most important IM3 products are $2f_1 - f_2$ and $2f_2 - f_1$. If they fall inside the passband, simple filtering cannot remove them without also affecting the wanted signal.

6. Third-Order Intercept (IP3 / TOI)

The third-order intercept point is an extrapolated linearity metric. It is the input- or output-referred power at which the extrapolated third-order products would equal the extrapolated fundamentals. It should be interpreted only in the small-signal region before compression.

$$IIP3 = P_{in} + (\Delta P / 2), \quad OIP3 = P_{out} + (\Delta P / 2), \quad \Delta P = P_{fund} - P_{IM3} \quad (5)$$

where ΔP is the decibel spacing between the fundamental and the third-order product. A higher IP3 denotes a more linear stage and a wider spur-free dynamic range. Figures 8 and 9 show the two-tone IP3 test bench, and Figure 10 the closely-spaced two-tone setup and the resulting intermodulation with tones 1 MHz from the carrier.

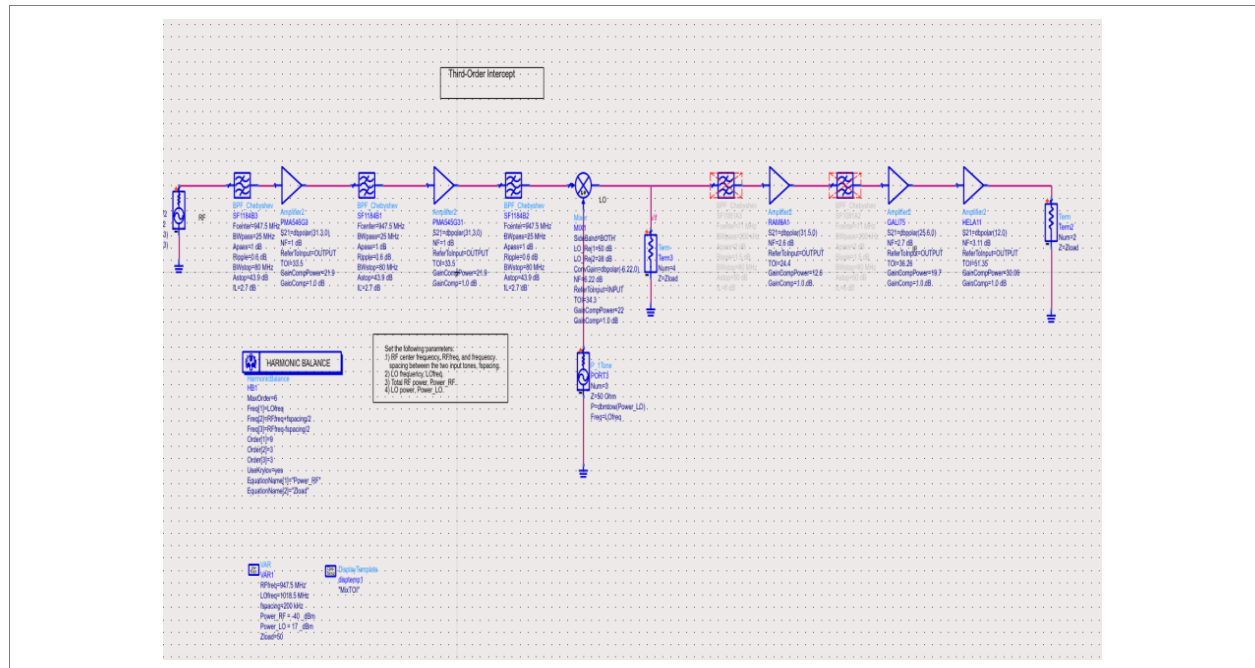


Figure 8. Two-tone third-order-intercept test bench.

The two-tone setup injects two nearby tones into the nonlinear receiver chain. The output spectrum is used to measure fundamental tones and third-order products, enabling input- and output-referred IP3 estimation.

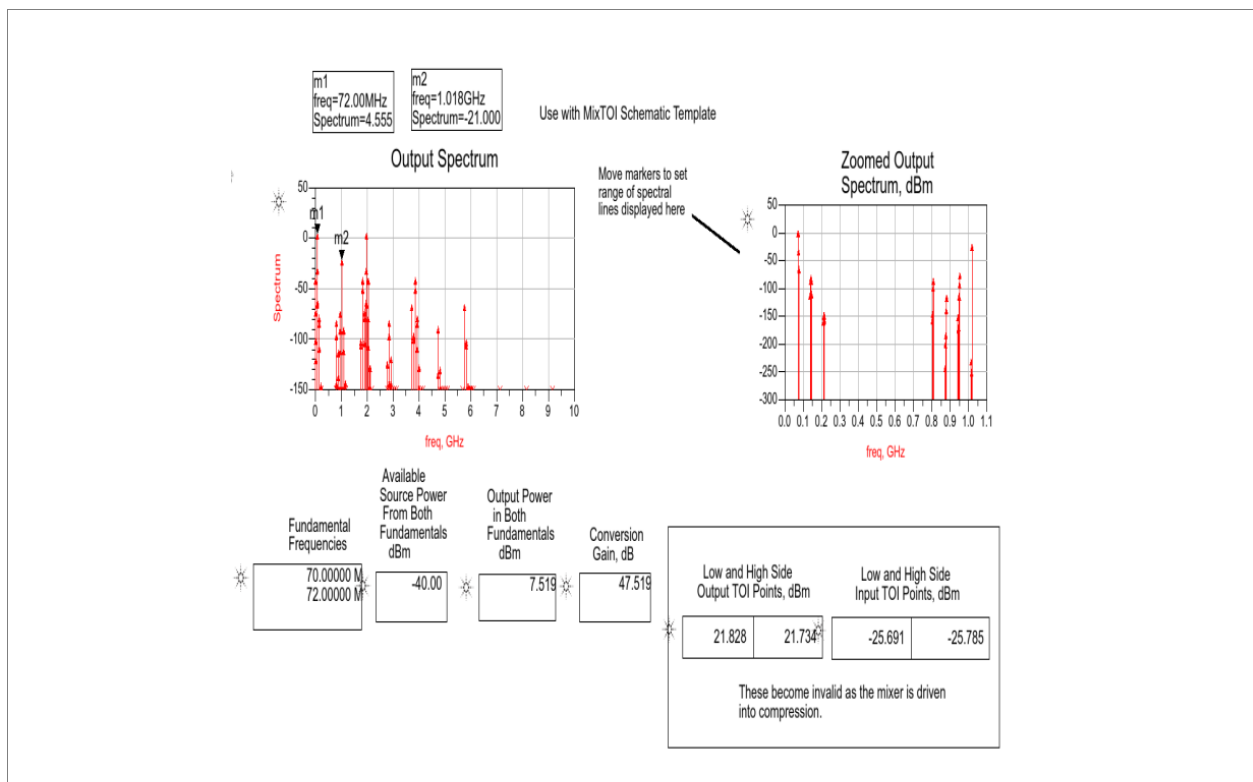


Figure 9. TOI result for tones at 70 MHz and 72 MHz.

The visible ADS marker table reports source power of -40 dBm, output power in both fundamentals of 7.519 dBm, conversion gain of 47.519 dB, OIP3 of 21.828/21.734 dBm and IIP3 of -25.691/-25.785 dBm. These values are valid only before the mixer is driven into compression, as noted in the ADS result panel.

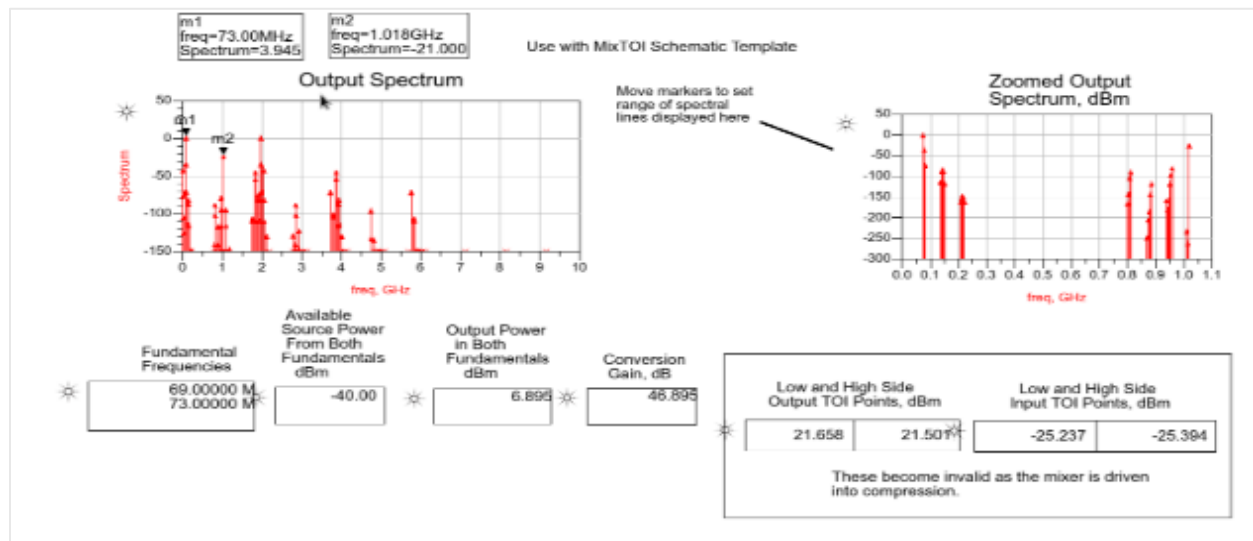


Figure 10. TOI result with tones at 69 MHz and 73 MHz.

The visible result shows output power in both fundamentals of 6.895 dBm, conversion gain of 46.895 dB, OIP3 of 21.658/21.507 dBm and IIP3 of -25.237/-25.394 dBm. This comparison shows how tone spacing and filtering influence the measured intermodulation result.

This result is consistent with the 70/72 MHz two-tone case. The reported values are Pout fundamentals 7.517 dBm, conversion gain 47.517 dB, OIP3 about 21.826/21.732 dBm and IIP3 about -25.692/-25.786 dBm. Filtering suppresses selected out-of-band products after generation; it does not improve the intrinsic linearity of the nonlinear device.

To demonstrate the benefit of channel filtering on intermodulation, the tone spacing was varied and an IF filter introduced. Figure 11 shows the response for the first tone 1 MHz from the carrier; Figures 12 and 13 compare the 1 MHz and 2 MHz spacings with the IF filter in place.

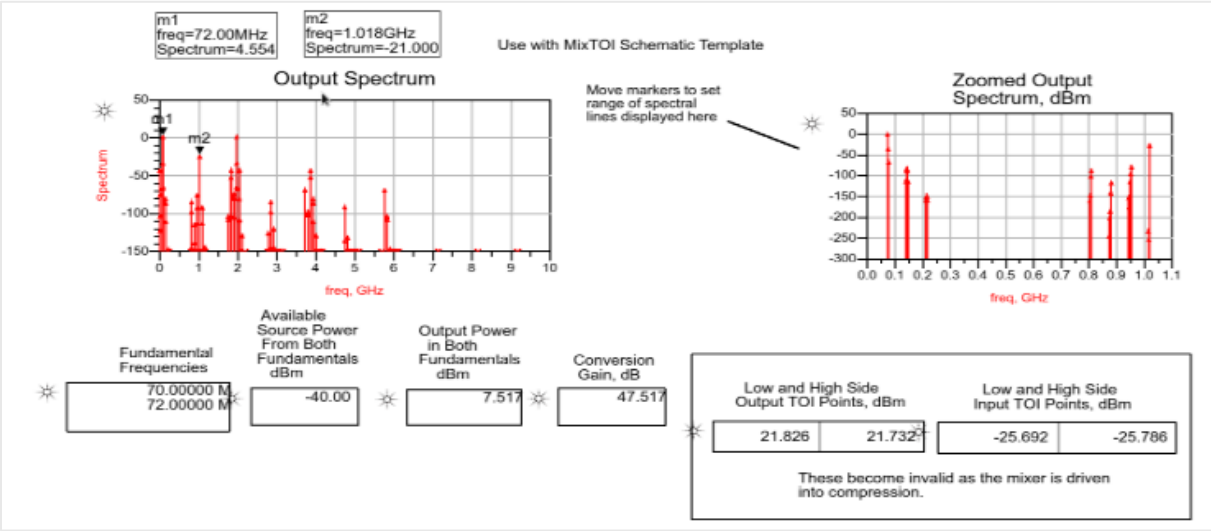


Figure 11. TOI result with tones at 70 MHz and 72 MHz after filtering condition.

Tone case	Input power	Pout fundamentals	Conversion gain	OIP3 low/high	IIP3 low/high
70/72 MHz	-40 dBm	7.519 dBm	47.519 dB	21.828 / 21.734 dBm	-25.691 / -25.785 dBm
69/73 MHz	-40 dBm	6.895 dBm	46.895 dB	21.658 / 21.507 dBm	-25.237 / -25.394 dBm
70/72 MHz filtered case	-40 dBm	7.517 dBm	47.517 dB	21.826 / 21.732 dBm	-25.692 / -25.786 dBm

Table 6. Parameter value

The IF filter suppresses out-of-band intermodulation products, and improves post-filter interference rejection. It does not make the nonlinear device itself more linear.

7. Transistor Bias Setup

The circuit-level amplifier design starts with DC biasing, which simultaneously sets the small-signal gain, the noise figure, and the linearity of the device; it must therefore be established and verified before any small-signal design. Bias determines transconductance, gain, noise figure, compression behavior and linearity. A bias point should therefore be selected before performing S-parameter, noise-circle or matching-network analysis.

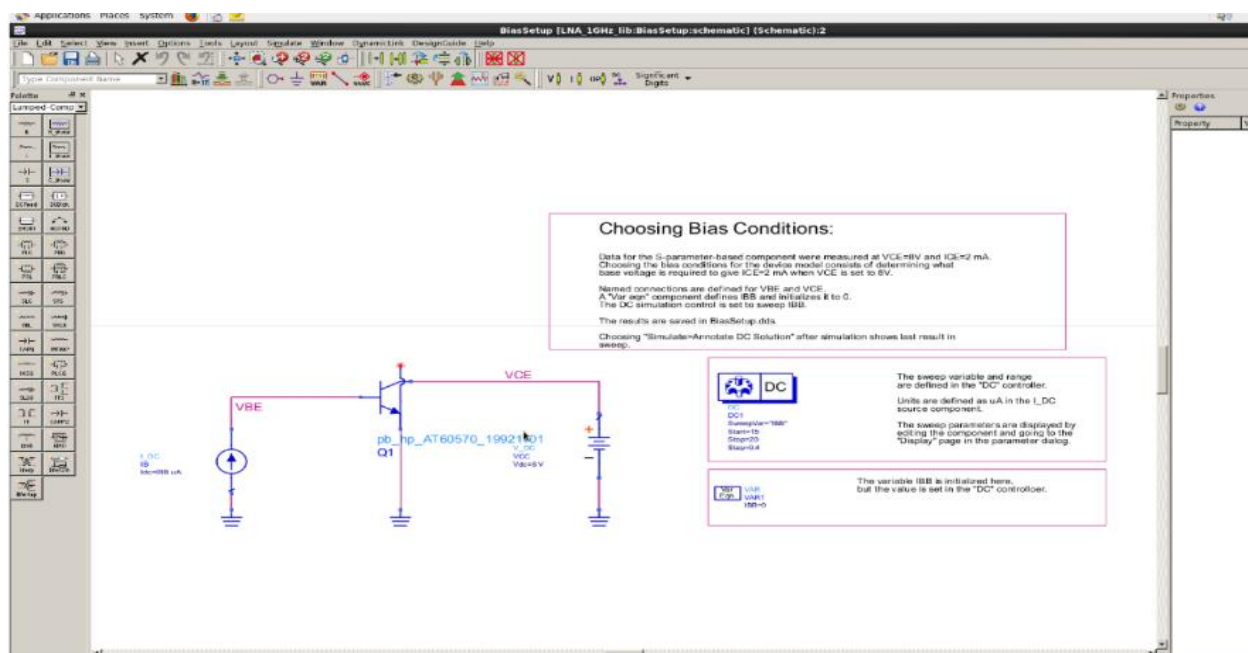


Figure 12. Bias-setup schematic

The schematic sweeps or evaluates the transistor bias network. The goal is to select a stable DC operating point that satisfies the required collector/base current, voltage and RF operating condition. The transistor bias network is evaluated to select the required current and voltage operating point.

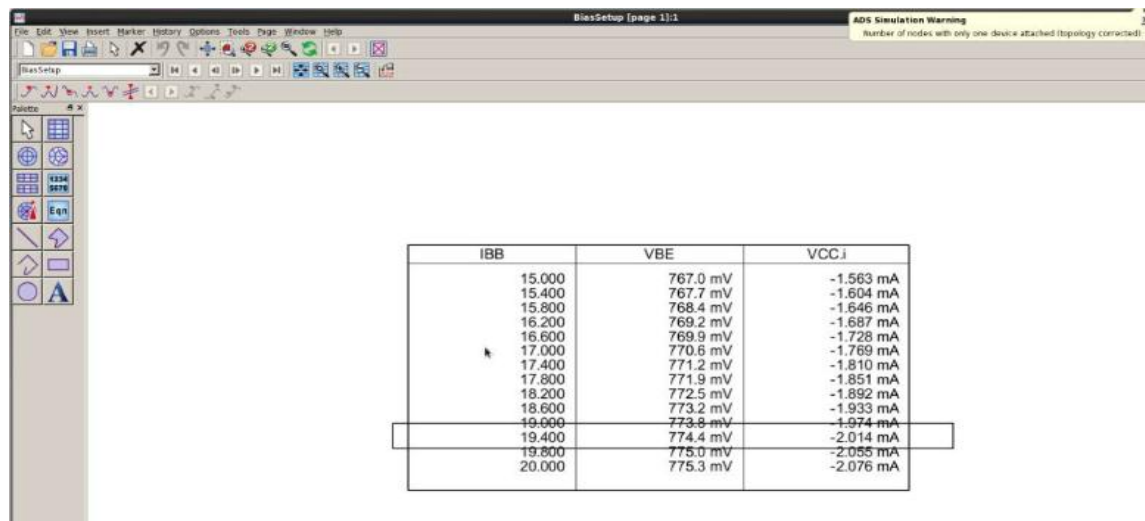


Figure 13. Bias-setup simulation table.

The highlighted row shows $IBB = 19.40$, $VBE = 774.4\text{ mV}$ and $VCC\text{ current} = -2.014\text{ mA}$. The negative current sign follows the ADS current-reference direction. The physical current magnitude is therefore approximately 2.014 mA .

Bias quantity	Value	Comment
Selected IBB	19.40	Highlighted ADS sweep row
Base-emitter voltage VBE	774.4 mV	Selected operating point
VCC current magnitude	about 2.014 mA	ADS table reports -2.014 mA due to sign convention
Design meaning	Bias point suitable for S-parameter/noise/matching evaluation	Bias must be stable before small-signal design

Table 6. Bias operating-point summary.

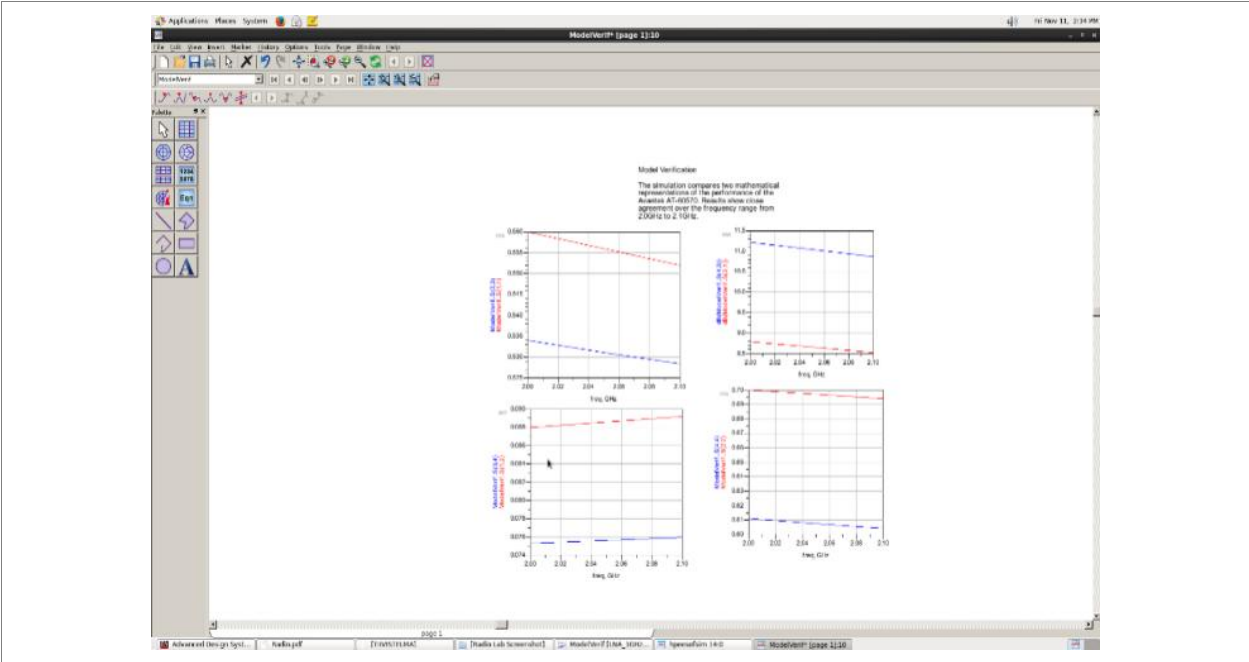


Figure 14. Bias-setup simulation result — verification of the DC operating point.

8. Model Verification and S-Parameters

Before a transistor model is trusted for design, its simulated S-parameters are compared against the manufacturer's measured data. The scattering parameters relate the incident and reflected travelling waves at the device ports; agreement on the Smith chart and in rectangular magnitude plots confirms the model is valid across the design band. Figure 14 shows the verification schematic and Figure 15 the Smith-chart comparison. S-parameters describe incident and reflected travelling waves at the RF ports and are the basis for gain, return-loss, stability, noise and matching analysis.

$$[b1 \ b2]^T = [S11 \ S12; S21 \ S22] [a1 \ a2]^T \text{-----}(6)$$

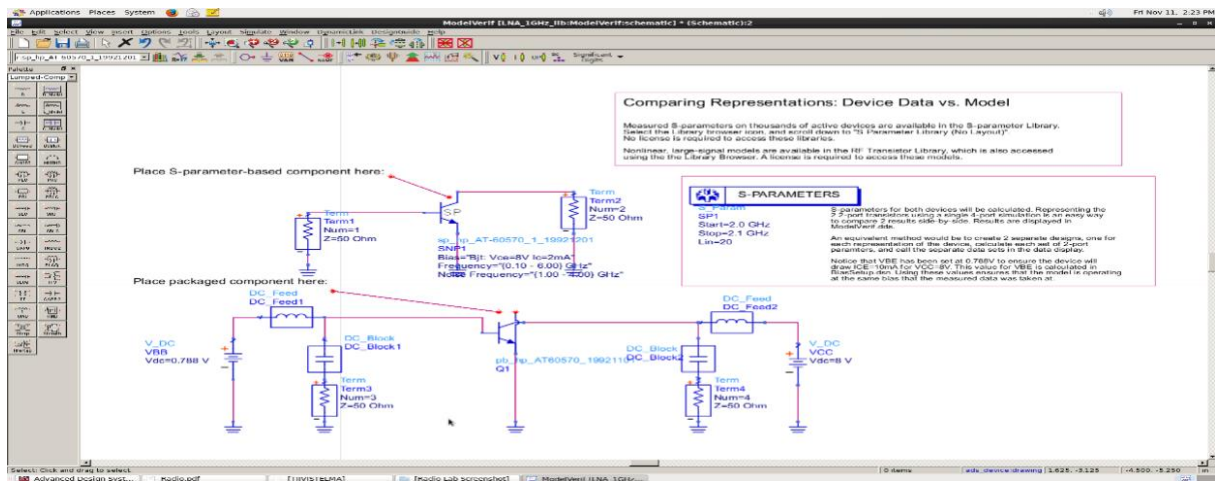


Figure 15. Model-verification schematic. The transistor model, biasing elements, and S-parameter controller are connected for small-signal verification.

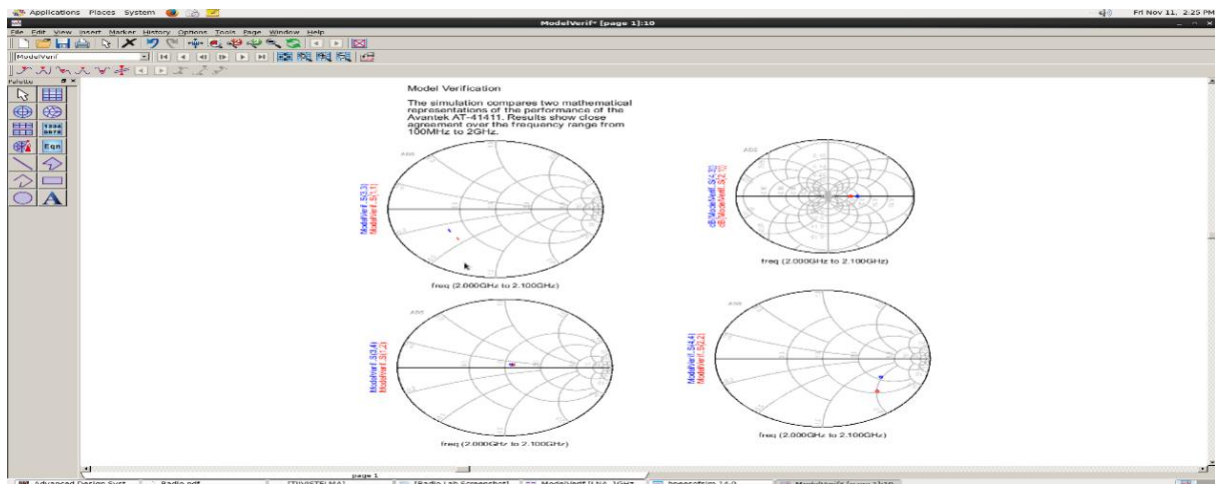


Figure 16. Model-verification schematic.

The Smith-chart comparison provides a visual check that the simulated model follows the reference impedance trajectory across the design band. This is a standard RF validation step before using a device model for matching-network design. The schematic connects the transistor model, biasing elements and S-parameter controller. The goal is to ensure the model behaves consistently under the selected bias and frequency range.

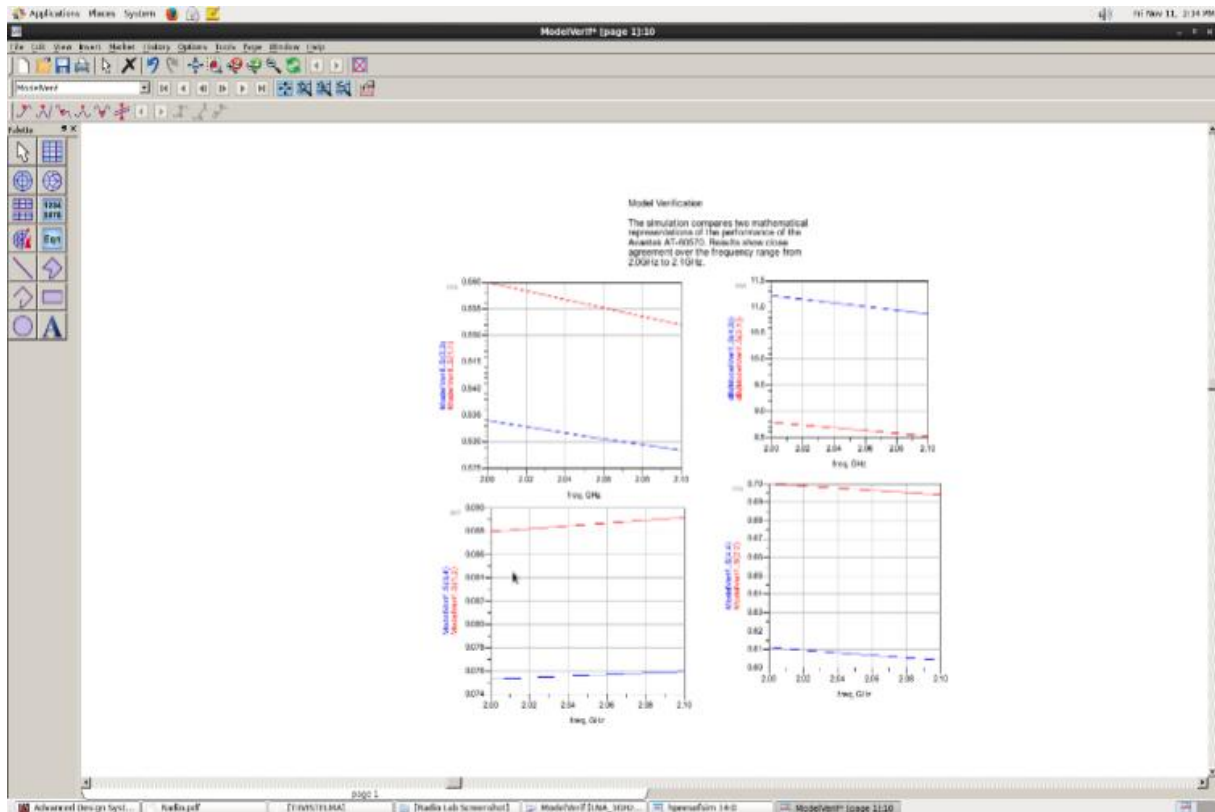


Figure 17. Rectangular S-parameter plots. Magnitude and phase trends provide a frequency-domain comparison of S11, S21, S12, and S22 behavior.

Rectangular magnitude/phase plots complement the Smith chart by showing frequency-dependent deviations more clearly. This helps verify S11, S21, S12 and S22 trends across the design band. Simulated versus reference S-parameter comparison. The plotted trends validate the model sufficiently for proceeding to noise and matching analysis. The overlaid trends show that the simulated model is sufficiently aligned with reference behavior for proceeding to noise, stability and matching design.

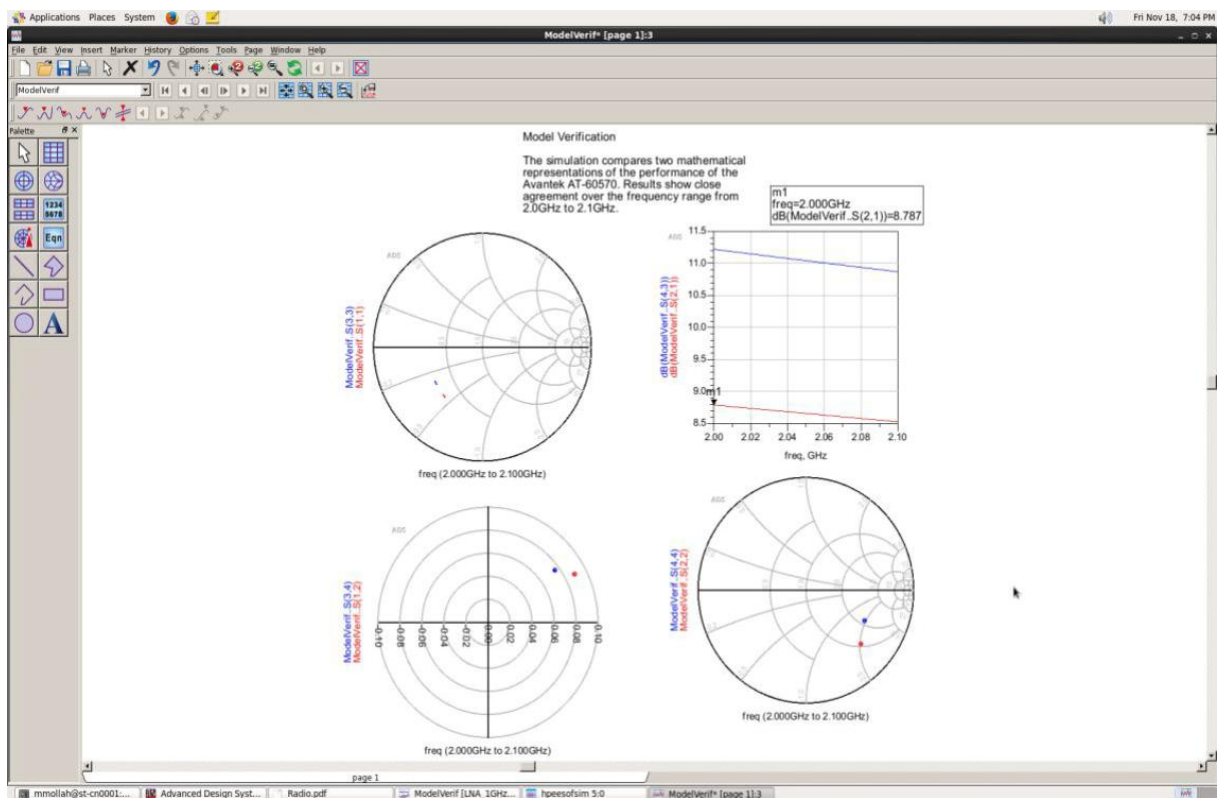


Figure 18. Combined S-parameter result plots. Smith-chart and rectangular representations jointly summarize input/output match, gain, and reverse isolation behavior.

9. Bias-Network Design, Noise, and Stability Analysis,

A practical RF amplifier must isolate the DC bias path from the RF signal path. Bias tees, RF chokes, bypass capacitors and stabilization components are used so that the transistor receives the required DC operating point while the RF ports still see the intended source and load impedances. With the model verified, a practical bias network is designed to deliver the operating point while presenting an RF-transparent path, decoupling the DC supply from the signal so that the matching networks see only the intended device impedance. Figure 19 shows the bias-network schematic and Figure 20 its simulated response.

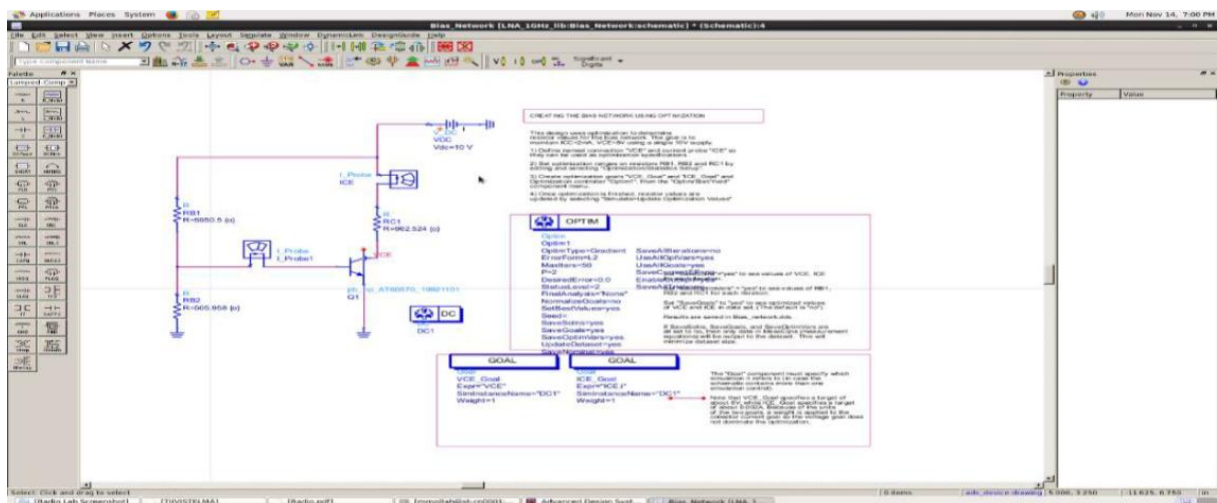


Figure 19. Bias-network schematic. The network provides the transistor operating point while separating RF and DC paths.

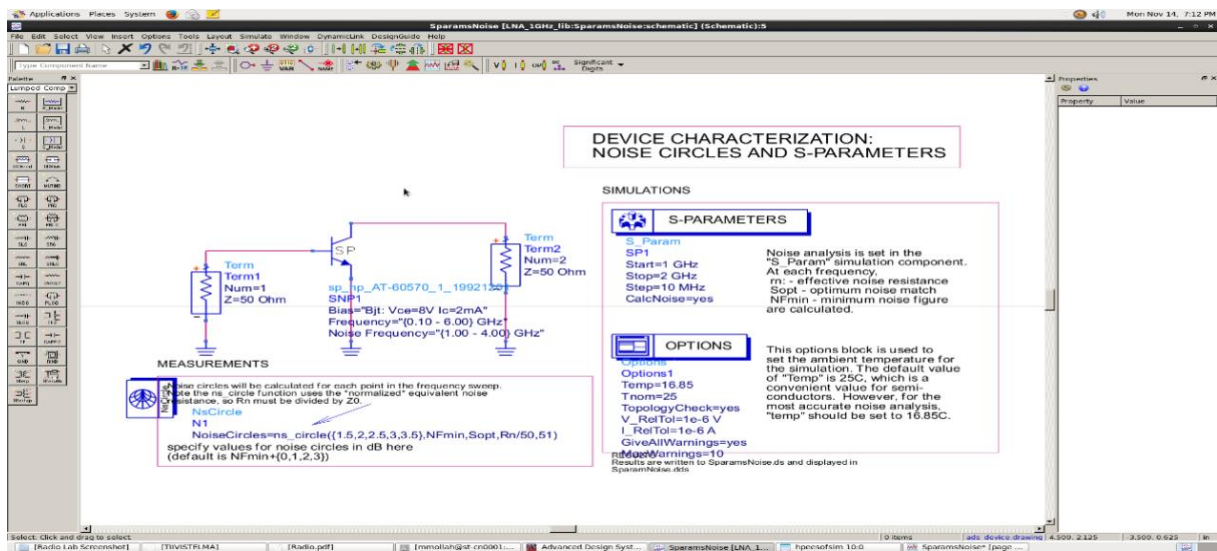


Figure 20. S-parameter/noise-analysis schematic. The setup prepares the circuit for S-parameter, noise, stability, and matching evaluation.

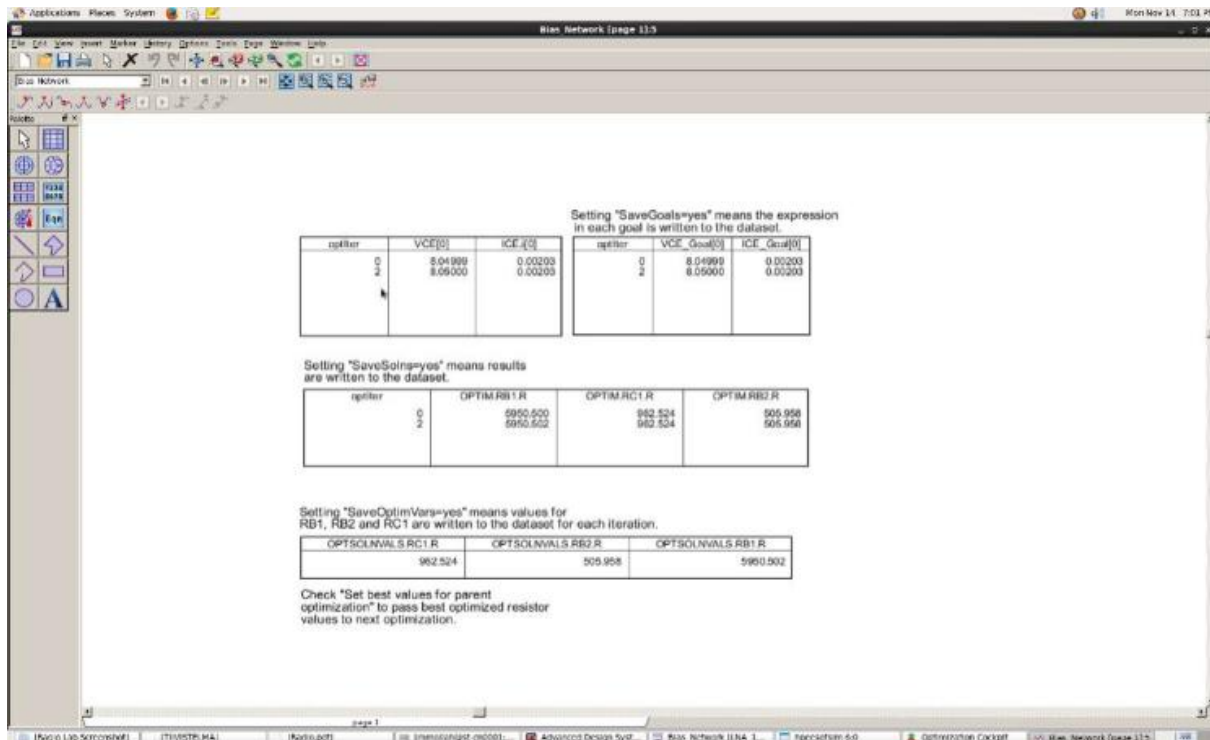


Figure 21. Bias-network simulation result.

The result confirms the bias-network behavior used before noise and matching analysis. For RF amplifier stability, Rollett stability should be evaluated using K and Delta, or alternatively the mu stability factor. The visible optimization table reports values such as VCE near 8.05 V and ICE near 0.00203 A for the optimized bias network, together with resistor values used in the bias network.

For RF amplifier stability, the Rollett stability factor and determinant should be evaluated. A common unconditional stability condition is $K > 1$ and $|\Delta| < 1$ over the frequency range of interest. The mu stability factor may also be used.

A common unconditional-stability condition is:

$$K = (1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2) / (2 |S_{12} S_{21}|) \text{ -----(7)}$$

Where, $K > 1$ and $|\Delta| < 1$; $\Delta = S_{11} S_{22} - S_{12} S_{21}$

10. Noise Figure, Optimum Source Match, and Noise Circles

An LNA is designed by trading off noise figure, gain, and stability. Minimum noise figure and maximum gain generally occur at different source impedances, so the source match is selected to balance the required performance targets.

$$F = F_{\min} + (4 R_n / Z_0) |\Gamma_S - \Gamma_{\text{opt}}|^2 / ((1 - |\Gamma_S|^2) |1 + \Gamma_{\text{opt}}|^2) \text{ -----(8)}$$

The schematic implements the RF/DC separation and prepares the circuit for noise and matching analysis. In practice, the RF path and bias path must be checked together because bias elements can introduce parasitic loading.

A low-noise amplifier is defined by the trade-off between minimum noise figure and maximum available gain; the two are obtained at different source impedances. Noise analysis identifies the source impedance that minimizes noise figure. For an LNA, the optimum noise match is often different from the conjugate gain match, so the design is a trade-off among NF, gain, stability and return loss.

For unconditional stability using the Rollett criterion, $K > 1$ and $|\Delta| < 1$ should be satisfied over the frequency range of interest. Alternatively, the stability factor $\mu > 1$ can be used.

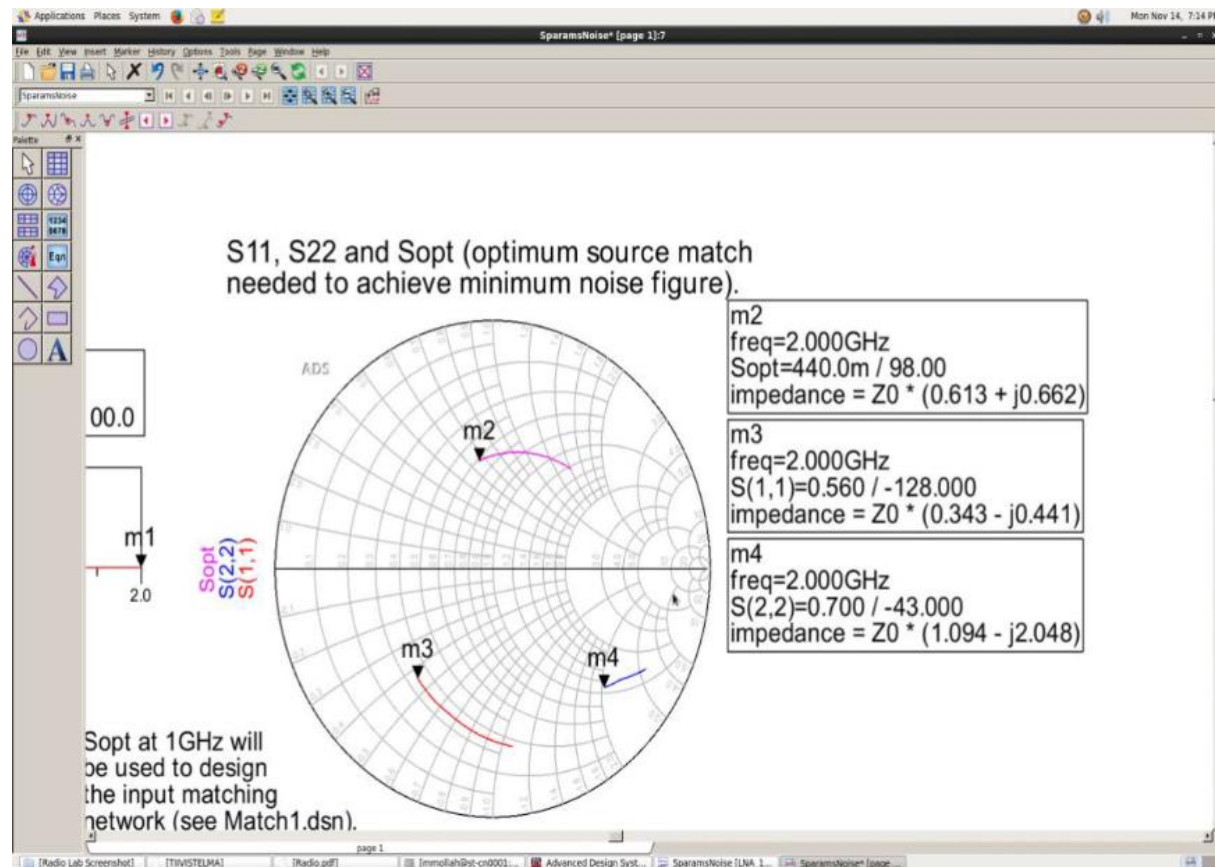


Figure 22 a. S11, S22 and optimum source match on the Smith chart.

At 2.0 GHz, the visible markers show $S_{opt} = 0.440$ angle 98 degrees with normalized impedance $0.613 + j0.662$, $S_{11} = 0.560$ angle -128 degrees with normalized impedance $0.343 - j0.441$, and $S_{22} = 0.700$ angle -43 degrees with normalized impedance $1.094 - j2.048$. These values guide source and load matching.

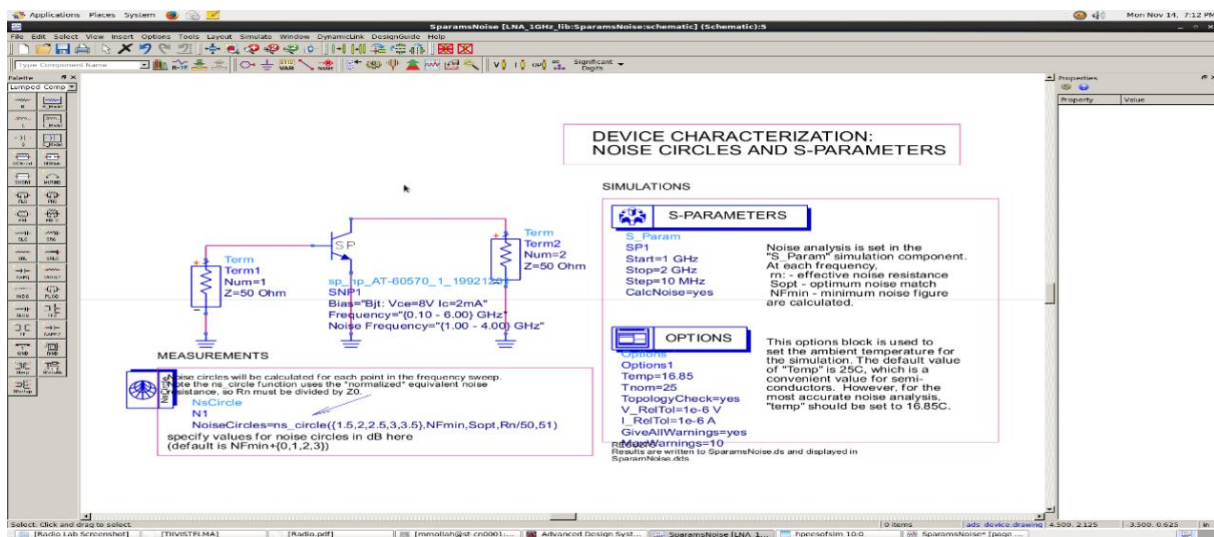


Figure 22 b. S-parameter / noise-analysis schematic.

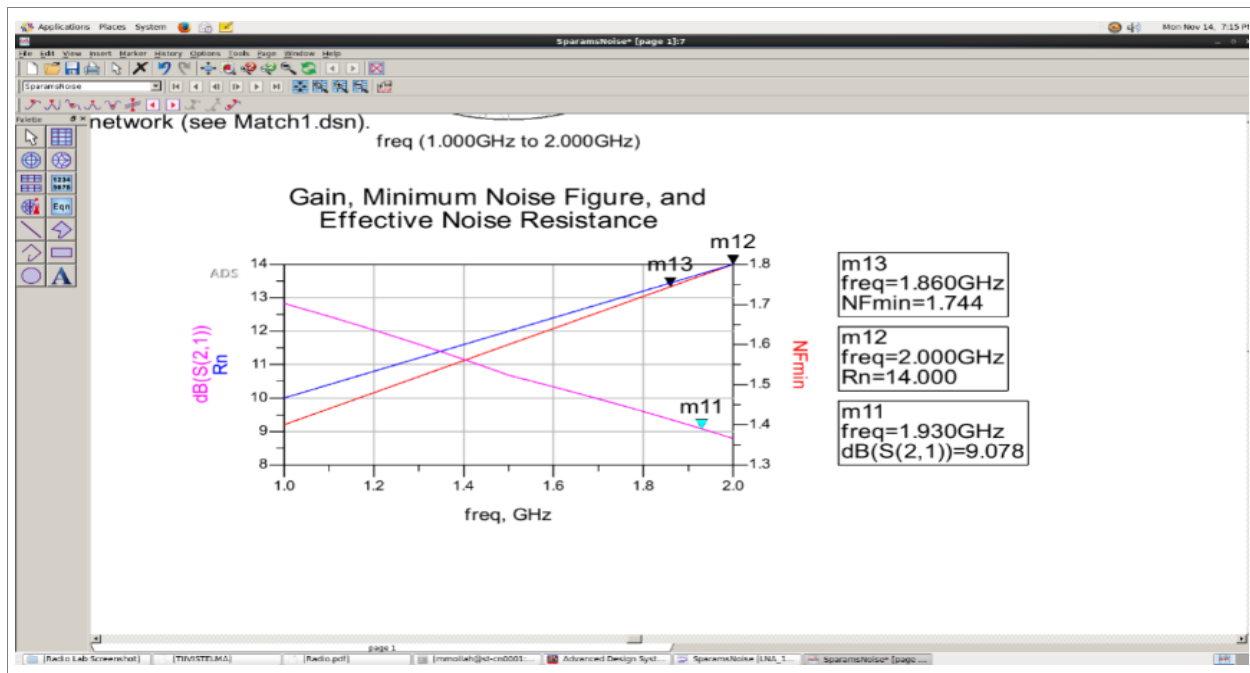


Figure 23. Gain, minimum noise figure and effective noise resistance.

The visible markers show $NFmin = 1.744$ at 1.86 GHz, $R_n = 14.000$ at 2.0 GHz and $dB(S_{21}) = 9.078$ at 1.93 GHz. These values are key circuit-level LNA design results.

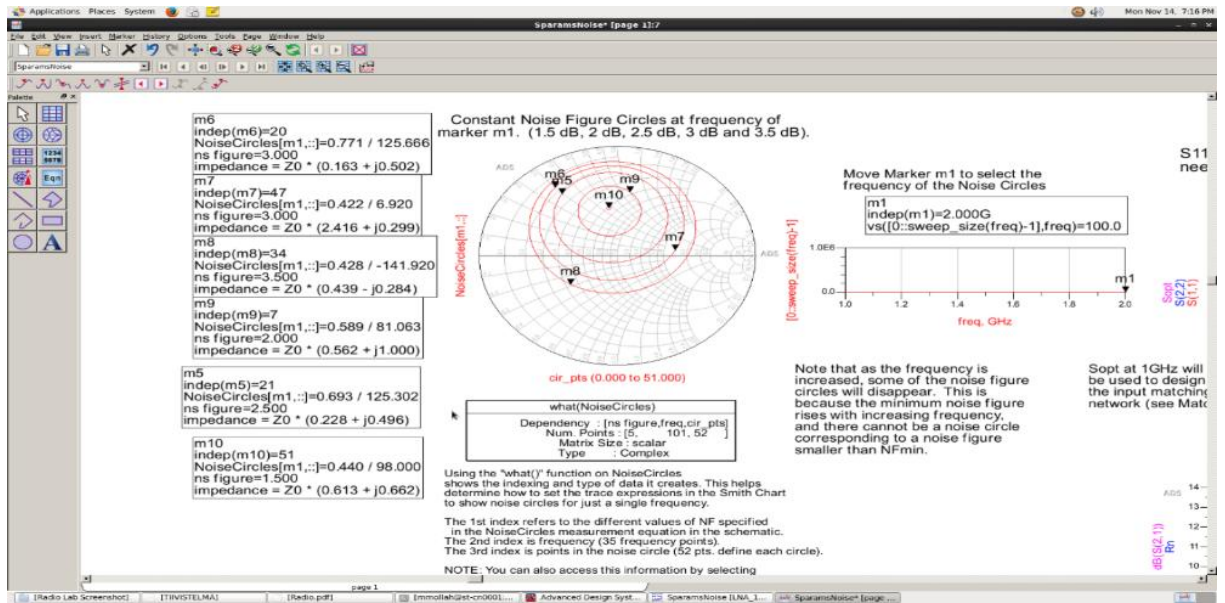


Figure 24. Constant noise-figure circles.

The noise circles show source impedances that produce specified noise figures at the selected frequency. The visible markers include 1.5 dB, 2.0 dB, 2.5 dB, 3.0 dB and 3.5 dB noise circles. This is used to choose a practical source impedance close to Γ_{opt} while balancing gain and stability.

Quantity	ADS value	Interpretation
Sopt at 2.0 GHz	0.440 angle 98 deg	Optimum source reflection coefficient for minimum noise.
Normalized Zopt	0.613 + j0.662	Normalized impedance 0.613 + j0.662 Equivalent source impedance target on Smith chart.
S11 at 2.0 GHz	0.560 angle -128 deg	Normalized impedance 0.343 - j0.441 Input reflection coefficient before final matching.
S22 at 2.0 GHz	0.700 angle -43 deg	Normalized impedance 1.094 - j2.048 Output reflection coefficient before final matching.
NFmin	1.744 at 1.86 GHz	Minimum noise figure marker from ADS plot.
Rn	14.000 at 2.0 GHz	Effective noise resistance marker from ADS plot.
Gain S21	9.078 dB at 1.93 GHz	Small-signal gain marker

Table 7. Noise, source-match, and S-parameter values extracted from ADS plots.

11 Matching-Network Synthesis and Final Amplifier

The final step transfers maximum power into the load by conjugately matching the device's output impedance. For a two-element (L-section) match, the inductor and capacitor values are synthesized from the loaded quality factor from which the reactances and hence the L/C values at the design frequency follow. The loaded quality factor and reactances are estimated from:

$$Q = \sqrt{(R_{high}/R_{low} - 1)} \text{-----(9)}$$

$$X_s = Q R_{low}, \quad X_p = R_{high}/Q \text{-----(10)}$$

$$L = X_L/(2\pi f), \quad C = 1/(2\pi f X_C) \text{-----(11)}$$

For a complex transistor impedance, the reactive part is first absorbed or cancelled, and the remaining real resistance is transformed to the required load using the matching network. ADS Smith-chart and tuning tools are used to synthesize and verify the final component values. Figure 26 shows the synthesized output matching network and its component values.

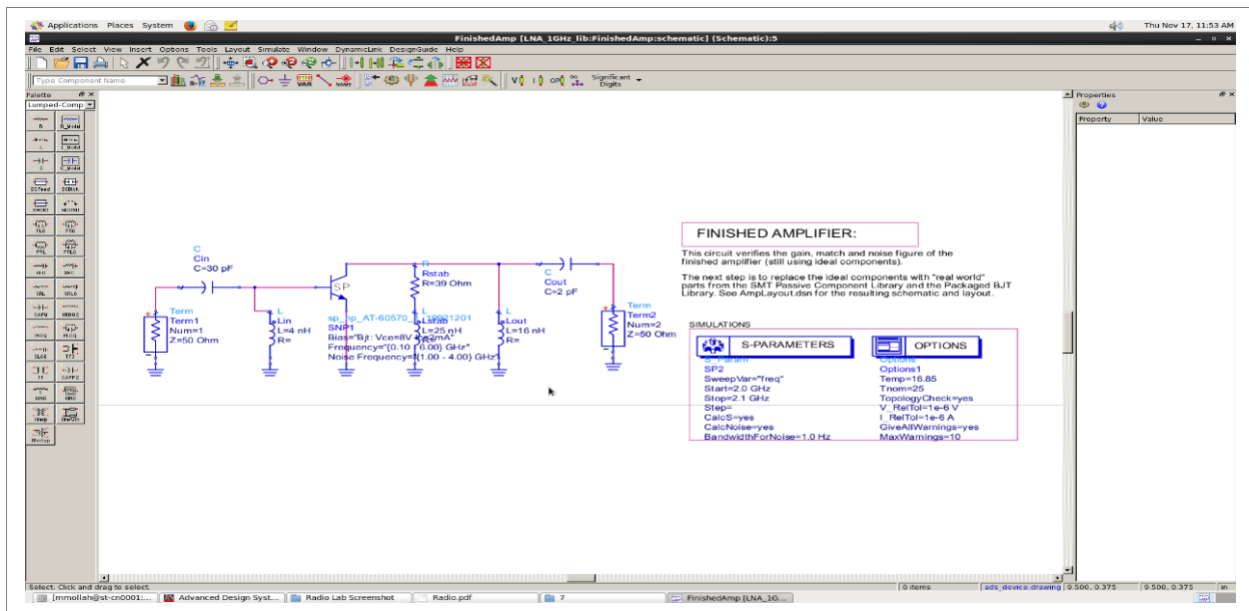


Figure 25. Output matching network- schematic and synthesized inductor/capacitor values.

The network includes matching and stabilization components used in the final amplifier design. The output match is added after the device has been biased, stabilized and source-matched. The visible schematic includes $C_{in} = 30$ pF, $L_{in} = 4$ nH, $R_{stab} = 39$ ohm, $L_{stab} = 25$ nH, $L_{out} = 16$ nH and a tunable/output capacitor path.

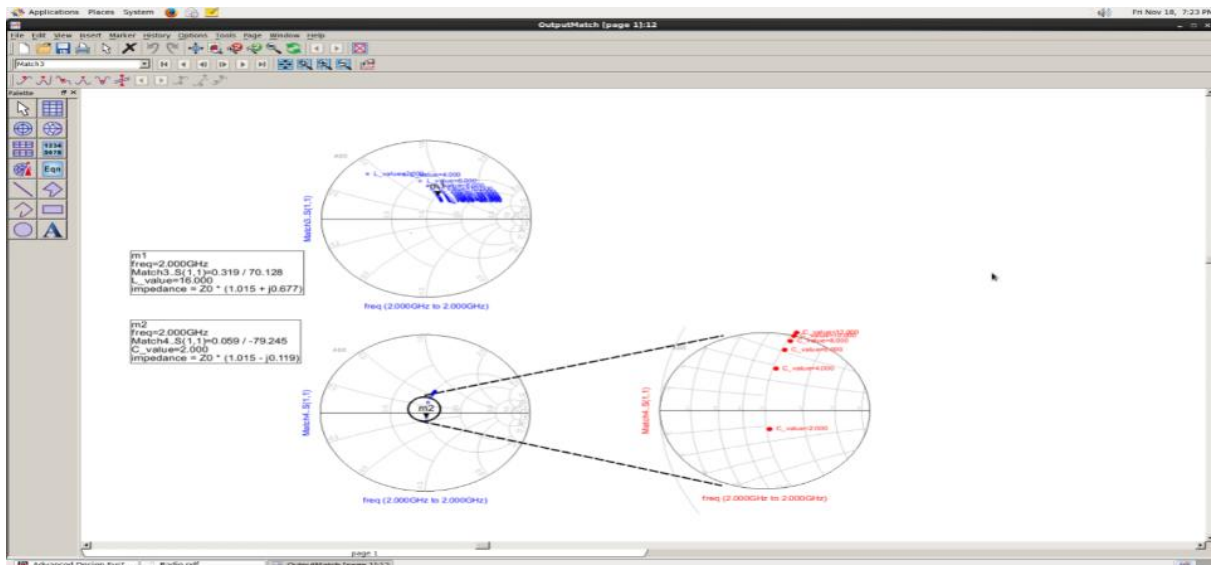


Figure 26. Determination of output matching L/C values. Smith-chart movement is used to translate the device output impedance to the target load impedance.

At 2.0 GHz, the visible markers show Match4_S11 = 0.319 angle 70.128 degrees with normalized impedance $1.015 + j0.677$ and Match4_S21 = 0.059 angle -79.245 degrees with normalized impedance $1.015 - j0.119$. The Smith-chart movement verifies impedance transformation toward the target region.

The final amplifier integrates the source match, active transistor, stabilization network, output match and 50-ohm ports. Visible values include $C_{in} = 30\text{ pF}$, $L_{in} = 4\text{ nH}$, $R_{stab} = 39\text{ ohm}$, $L_{stab} = 25\text{ nH}$, $L_{out} = 16\text{ nH}$ and $C_{out} = 2\text{ pF}$. This completes the transition from RF system analysis to a realizable matched amplifier circuit.

Component	Final value	Role
Cin	30 pF	Input coupling / matching capacitor
Lin	4 nH	Input matching inductor
Rstab	39 ohm	Stabilization resistor
Lstab	25 nH	Stabilization / RF path inductance
Lout	16 nH	Output matching inductor
Cout	2 pF	Output coupling / matching capacitor
Ports	50 ohm source and load	Standard RF system impedance

Table 8. Final matching and stabilization component values extracted from the ADS design screenshots.

12. Skills and Competencies Demonstrated

This project demonstrates a complete RF and microwave engineering workflow, starting from system-level RF performance analysis and progressing toward transistor-level amplifier realization in Keysight ADS. It shows the ability to connect RF system requirements, modulation quality, receiver dynamic range, device characterization, noise optimization, stability analysis, and impedance-matching design within one coherent engineering flow.

At the RF system level, the project demonstrates practical capability in ACPR/ACLR analysis, EVM interpretation, receiver link-budget construction, spurious-response evaluation, and two-tone IP3 extraction. These tasks show understanding of how nonlinear RF behavior affects both spectral emissions and wanted-signal quality. The ACPR and EVM analysis connects amplifier distortion to adjacent-channel leakage and constellation degradation, while the receiver budget links gain, noise figure, and dynamic-range allocation across cascaded RF, mixer, IF, and baseband stages.

At the circuit-simulation level, the project demonstrates hands-on ADS capability in transistor biasing, S-parameter simulation, device-model verification, noise-figure analysis, stability assessment, Smith-chart interpretation, and input/output matching-network synthesis. The biasing work confirms the DC operating point before small-signal design, while the S-parameter verification establishes whether the transistor model is suitable for further RF design. The noise and stability analysis show the ability to evaluate (NF_{min}) , (Γ_{opt}) , (R_n) , (S_{11}) , (S_{21}) , (S_{22}) , Rollett stability factor (K), and matching trade-offs.

The project also demonstrates strong RF linearity understanding through two-tone intermodulation analysis, IIP3/OIP3 calculation, compression-limit awareness, and interpretation of third-order products at $(2f_1 - f_2)$ and $(2f_2 - f_1)$. The IF-filtering analysis is treated correctly: filtering suppresses selected out-of-band products after they are generated and improves post-filter interference rejection, but it does not improve the intrinsic linearity of the nonlinear device itself.

The noise and sensitivity analysis demonstrates understanding of Friis cascade theory and the importance of early-stage RF performance. The project applies the principle that front-end loss and early LNA gain strongly influence the overall receiver noise figure. It also shows the ability to interpret noise circles and select a practical source impedance by balancing minimum noise figure, gain, stability, and realizable matching-network constraints.

The impedance-matching work demonstrates RF design capability using Smith charts, reflection coefficients, conjugate matching principles, and L-section matching equations. The final matching network summarizes the input and output L/C elements together with stabilization components, showing the connection between theoretical impedance transformation and practical RF circuit implementation.

Beyond simulation, the project demonstrates disciplined engineering communication. Each schematic and result is interpreted technically, equations are traceable to the design task, and numerical values are clearly identified as either ADS marker values or approximate values extracted from visible plots. This distinction makes the report credible, reviewable, and defensible in a technical interview.

Overall, this project demonstrates the capability to analyze, simulate, explain, and document a complete RF system and amplifier design workflow. It highlights practical competence in RF system design, RF circuit simulation, linearity analysis, receiver budgeting, transistor-level amplifier design, noise/stability trade-offs, and professional technical documentation.

13. Conclusion

A complete RF design study was carried out in Keysight ADS, spanning system-level characterization and circuit-level amplifier design. At the system level, transmitter linearity and signal quality were quantified through ACPR and EVM under $\pi/4$ -DQPSK modulation; a cascaded receiver link-budget established sensitivity and dynamic range; and spurious-response and two-tone IP3 analyses characterized intermodulation, with IF filtering shown to recover linearity headroom for closely-spaced interferers. The receiver side develops a cascaded link budget, spurious-response map and two-tone IP3 analysis. At the circuit level, a transistor amplifier was biased and its model verified against measured S-parameters, after which noise-circle and stability analysis guided the source match ($NF_{min} \approx 1.74$ near 1.86 GHz) and an L-section network matched the output. The work demonstrates command of the full RF design arc- from system budgets and modulated-signal linearity to a biased,

verified, noise-optimized, and impedance-matched amplifier- the core competency set for RF / microwave and PHY-layer hardware design.

14. References

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